

# Flash Flood Risk Intelligence Platform — Architecture Document

**Category:** Software (Disaster Management) **Scope:** Hackathon MVP ("Low-End") → Production-grade system ("High-End")

## 1. Solution Structure Overview

The system is organized into five logical layers, kept identical in principle between the low-end and high-end versions — only the *implementation depth* of each layer changes.

**Data Acquisition Layer** — weather, rainfall, terrain, historical disaster data, IoT sensors

**Ingestion Layer** — APIs, message brokers, schedulers

**Storage Layer** — relational + spatial + time-series data

**Risk Processing Layer** — rule-based engine (+ ML where justified)

**Application Layer** — map, dashboards, alerts

This consistency matters for judges: it shows the MVP is a *subset* of a coherent real system, not a toy that would need a rewrite to scale.

## 2. Low-End Architecture (Hackathon MVP)

### 2.1 Design Principles

Minimize moving parts that can fail live on stage.

One live data path (physical prototype → dashboard); rest of the grid is simulated but clearly labeled as such.

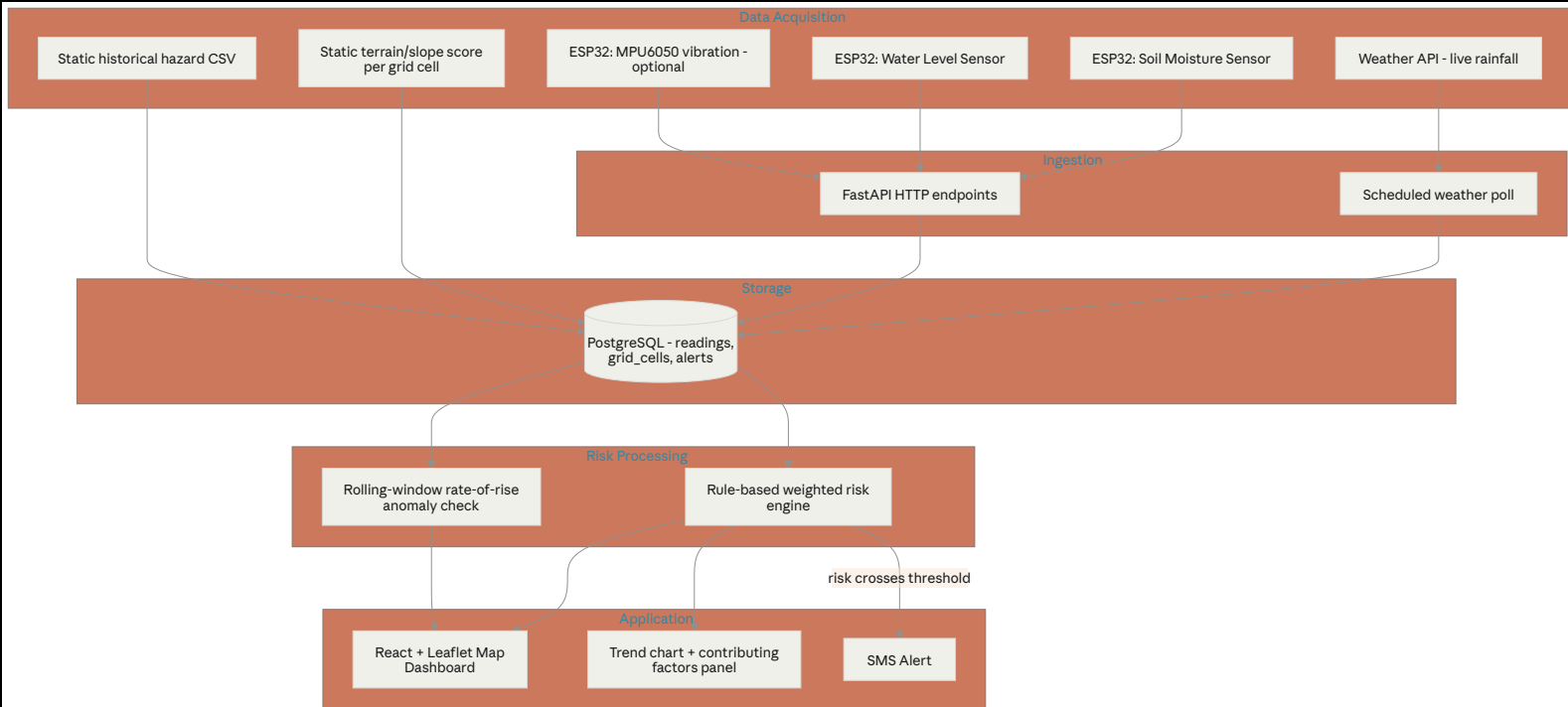
No unlabeled ML. No fabricated probabilities or lead times.

Everything explainable in one sentence to a judge.

### 2.2 Components

| Layer       | Component       | Choice                                                                              | Why                                                      |
|-------------|-----------------|-------------------------------------------------------------------------------------|----------------------------------------------------------|
| Acquisition | Rainfall        | Live public weather API (e.g., OpenWeather)                                         | Real, free, reliable                                     |
| Acquisition | Soil moisture   | 1x capacitive sensor on ESP32                                                       | Cheap, avoids resistive corrosion drift                  |
| Acquisition | Water level     | 1x ultrasonic (HC-SR04) or float sensor                                             | Enables rate-of-rise calc                                |
| Acquisition | Ground movement | MPU6050 (optional, relabeled "vibration anomaly")                                   | Demo-only, not slope stability                           |
| Acquisition | Terrain/slope   | Precomputed static score per grid cell from open elevation data (SRTM/Bhuvan)       | No live GIS engine needed                                |
| Acquisition | Historical      | Small static CSV of known landslide/flood-prone zones (GSI Bhukosh where available) | Adds legitimacy without a live pipeline                  |
| Ingestion   | IoT → Backend   | ESP32 → HTTP POST (skip MQTT broker for MVP)                                        | Fewer failure points live                                |
| Ingestion   | Scheduler       | Simple cron/async job polling weather API every few min                             | Lightweight                                              |
| Storage     | Database        | Single PostgreSQL instance (no PostGIS)                                             | Grid math via lat/lon bounding boxes is enough           |
| Processing  | Risk Engine     | Transparent weighted rule engine                                                    | Explainable, defensible under questioning                |
| Processing  | Anomaly layer   | Rolling-window threshold on rate-of-rise (statistical, not ML)                      | Legitimate "smart" feature without needing training data |
| Application | Map             | React + Leaflet, 4–6 hardcoded grid cells                                           | Reliable, no API quota risk                              |
| Application | Alerts          | 1 real channel (SMS via free-tier API, e.g., Twilio/MSG91 trial)                    | One well-tested channel beats four flaky ones            |

## 2.3 Low-End Architecture Diagram



## 2.4 What This Deliberately Excludes

- No trained ML model (explicitly stated as roadmap, not current capability)
- No MQTT broker, no PostGIS, no time-series DB — reduces demo failure surface
- No multi-channel alerting — one channel done well
- No claimed lead-time number in minutes/hours

## 3. High-End Architecture (Production / Pilot-Ready)

### 3.1 Design Principles

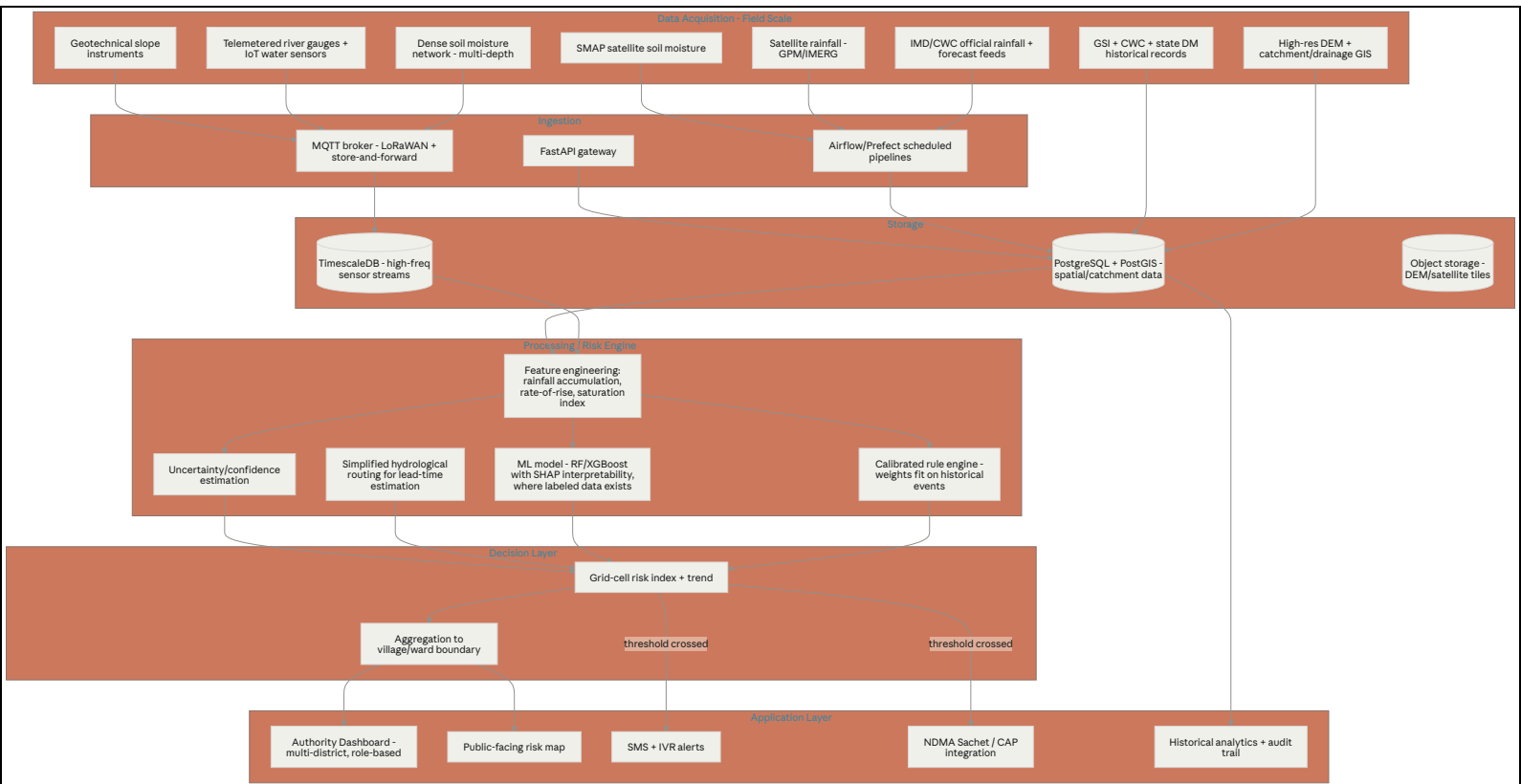
- Built for real deployment across many villages/wards with a real disaster management authority.
- Adds ML **only where labeled ground-truth data exists** via formal data partnerships (GSI, CWC, IMD, state DM authority).
- Resilient to field conditions: intermittent connectivity, sensor drift, scale.

### 3.2 Components

| Layer       | Component                          |  | Choice                                                                                                        | Why                                                          |
|-------------|------------------------------------|--|---------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|
| Acquisition | Rainfall/forecast                  |  | IMD/CWC official feeds + satellite rainfall estimates (e.g., GPM/IMERG)                                       | Regulatory-grade + spatial coverage beyond point stations    |
| Acquisition | Soil moisture                      |  | Dense in-situ sensor network (multi-depth: 15/30/60cm) + SMAP satellite soil moisture for coverage gaps       | Point sensors alone don't represent a ward                   |
| Acquisition | Water level                        |  | Telemetered river gauges + IoT ultrasonic/radar sensors at vulnerable channels                                | Redundant, higher accuracy                                   |
| Acquisition | Slope stability                    |  | Real geotechnical instruments: inclinometers, piezometers, extensometers (not MPU6050)                        | Scientifically valid slope monitoring                        |
| Acquisition | Terrain/GIS                        |  | High-res DEM, catchment delineation, land-use, drainage network                                               | Enables real hydrological routing                            |
| Acquisition | Historical                         |  | Full GSI Bhukosh + CWC flood records + state DM incident reports                                              | Real labels for ML                                           |
| Ingestion   | IoT                                |  | MQTT broker (e.g., EMQX) with store-and-forward + LoRaWAN for remote hilly connectivity                       | Handles field connectivity failures                          |
| Ingestion   | Data pipeline                      |  | Airflow/Prefect scheduled + streaming jobs                                                                    | Reliable, auditable data pipeline                            |
| Storage     | Spatial DB                         |  | PostgreSQL + PostGIS                                                                                          | Real geospatial queries, catchment/grid overlays             |
| Storage     | Time-series                        |  | TimescaleDB or InfluxDB                                                                                       | High-frequency sensor data at scale                          |
| Storage     | Object storage                     |  | S3-compatible, for satellite imagery/DEM tiles                                                                | Large geospatial assets                                      |
| Processing  | Risk Engine v1                     |  | Calibrated weighted rule engine (weights fit via logistic regression on historical events, not hand-assigned) | Bridges rule-based transparency with data-driven calibration |
| Processing  | Risk Engine v2 (where data allows) |  | Random Forest / XGBoost, precision/recall optimized, interpretable via SHAP                                   | Only where labeled data partnership exists                   |
| Processing  | Hydrological layer                 |  | Basic catchment rainfall-runoff routing (e.g., simplified SCS-CN or similar)                                  | Only real way to approach a defensible lead-time estimate    |
| Decision    | Uncertainty layer                  |  | Confidence bands, not fake single-number                                                                      | Scientific honesty at scale                                  |

|             |                     |                                                                                                                           |                                                    |
|-------------|---------------------|---------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|
| Decision    | Uncertainty layer   | probabilities                                                                                                             | Scientific honesty at scale                        |
| Application | Authority dashboard | Role-based access, multi-district view, audit trail                                                                       | Real operational tool for DM authorities           |
| Application | Alerts              | SMS + IVR (for low-literacy/no-smartphone areas) + Common Alerting Protocol (CAP) integration with NDMA's Sachet platform | Matches actual Indian govt alerting infrastructure |
| Ops         | Monitoring          | Sensor health monitoring, data-quality checks, alert-fatigue tracking                                                     | Required for trust at scale                        |

### 3.3 High-End Architecture Diagram



### 3.4 Key Upgrades vs Low-End

Point sensors → sensor **networks** + satellite data fusion (solves the "one sensor can't represent a village" problem)

Hand-assigned weights → **calibrated** weights, and real ML only where labeled data exists

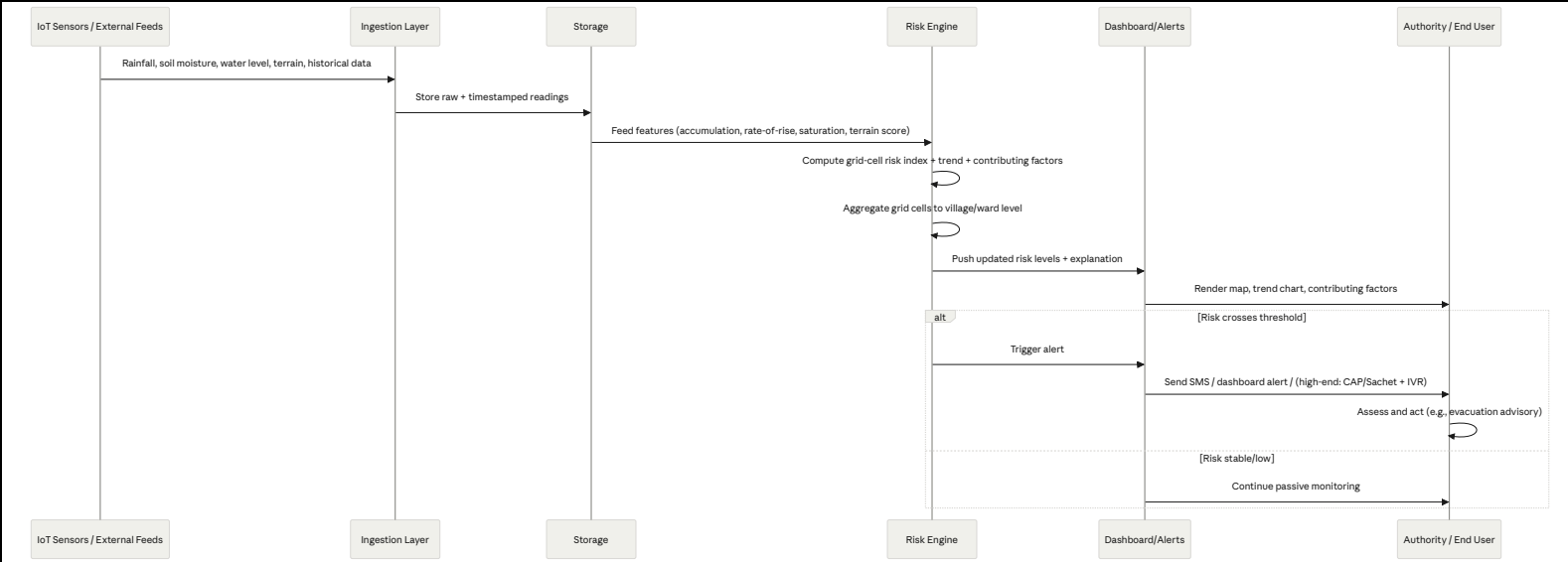
No lead-time claim → **hydrological routing model** as the only scientifically legitimate path to an actual time estimate

No connectivity resilience → **MQTT + LoRaWAN + store-and-forward** for real hilly-terrain conditions

Single alert channel → **CAP/Sachet integration**, the actual protocol Indian disaster alerting uses

Fake probability → **confidence bands**, explicitly presented as uncertainty, not false precision

## 4. Complete End-to-End Flow (applies to both versions, differing only in depth)



## In plain language:

Data comes in from live sources (weather, sensors) and static sources (terrain, history).

It's stored with timestamps and location tags.

The risk engine computes a transparent, explainable risk score per grid cell — and explicitly states *why* (contributing factors), not just *what*.

Grid-cell scores are aggregated up to human-readable village/ward units.

The dashboard shows current state + trend, not a false single-number prediction.

When risk crosses a defined threshold, an alert fires through a real channel.

Humans (authorities) make the evacuation decision — the system supports, not replaces, decision-making.

## 5. Migration Path: Low-End → High-End

| Step | What Changes                                           | Trigger                                                             |
|------|--------------------------------------------------------|---------------------------------------------------------------------|
| 1    | Add PostGIS, migrate grid math to real spatial queries | When cell count grows beyond a handful                              |
| 2    | Add MQTT + LoRaWAN                                     | When deploying sensors in real connectivity-poor terrain            |
| 3    | Add TimescaleDB                                        | When sensor read frequency/volume grows                             |
| 4    | Calibrate rule-engine weights via logistic regression  | Once even a small labeled historical dataset is secured             |
| 5    | Introduce ML (RF/XGBoost)                              | Only after formal data partnership with GSI/CWC/state DM authority  |
| 6    | Add hydrological routing for lead-time                 | Only with catchment-level DEM + rainfall-runoff modeling capability |
| 7    | Integrate with NDMA Sachet/CAP                         | At pilot-deployment stage with government partner                   |

This phased path is itself a strong talking point for judges — it shows the low-end system isn't a dead-end prototype but a genuine first step of the high-end one.